

COURSE TITLE : INORGANIC CHEMICAL TECHNOLOGY
COURSE CODE : 4073
COURSE CATEGORY : B
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 56
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Manufacturing of Soda Ash	14
2	Manufacturing procedures of A c i d s	14
3	Manufacturing of Ammonia and Ammonium sulphate Fertilizer	14
4	Manufacturing of pigments and Silicate	14
TOTAL		56

COURSE OUTCOME :

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	Understand the manufacturing methods of inorganic Chemicals.
2	2	Understand the manufacturing of Different Acids .
3	3	Understand the manufacturing of Fertilizers
	4	Understand the manufacturing of Pigments and Paints
	5	Comprehend the manufacturing methods of Silicate industries.

SPECIFIC OUTCOMES

MODULE – I

1.1.0 Understand the Manufacturing of Soda Ash

- 1.1.1 Explain the steps involved with preparation and purification of Brine
- 1.1.2 Explain the process of ammoniation of Brine
- 1.1.3 Explain the process of carbonation of ammoniated brine
- 1.1.4 Explain the construction and working of carbonating tower
- 1.1.5 Explain the reaction involved in the carbonating tower
- 1.1.6 Draw the sketch of carbonating towers
- 1.1.7 Explain the working of filters for filter bicarbonate
- 1.1.8 Explain the conversion of bicarbonate to carbonate
- 1.1.9 Explain the process for recovery of ammonia
- 1.1.10 Draw the flow diagram for the entire process of Solvay and modified Solvay process

1.2.0 Understand the manufacturing of Chlorine.

- 1.2.1 Explain the process manufacturing of chlorine.
- 1.2.2 Explain the chlorine compression and liquefaction

1.3.0 Understand manufacture of caustic soda.

- 1.3.1 Describe the membrane process of caustic soda
- 1.3.2 Explain the cathodic and anodic reaction

- 1.3.3 Explain the characteristic of membrane
- 1.3.4 Proportion of Flemion membrane
- 1.3.5 Explain the cell components
- 1.3.6 Explain caustic soda flake production

1.4.0 Understand the manufacturing of Aluminum Sulphate

- 1.4.1 Explain the manufacturing process of Aluminum Sulphate from Bauxite.

1.5.0. Understand the production of Lime

- 1.5.1. Explain the process of manufacture of lime from limestone.
- 1.5.2 Explain the production of Bleaching powder from lime

MODULE – II

2.1.0 Understand the manufacturing procedures of A c i d s

- 2.1.1 List the properties – Physical and Chemical
- 2.1.2 List the uses of Sulphuric acid
- 2.1.3 List the raw material for the manufacture of Sulphuric acid
- 2.1.4 Explain the reactions involved and heat evolved with the help of chemical equation
- 2.1.5 Explain cooling and cleaning of SO₂
- 2.1.6 Explain the working of waste heat boilers
- 2.1.7 Explain the general principles and theory behind contact process
- 2.1.8 Distinguish the advantages and disadvantages of platinum and Vanadium pentoxide catalyst
- 2.1.9 Explain the phenomenon of catalyst poisoning
- 2.1.10 Explain the catalyst carriers
- 2.1.11 Explain the constructional details and working of converters
- 2.1.12 Explain the principles of absorption of SO₃
- 2.1.13 Explain the constructional details and working of SO₃ absorbers
- 2.1.14 Explain the production method of oleum
- 2.1.15 Explain the principles of DCDA process with flow sheet

2.2.0 Understand the production of Nitric acid

- 2.2.1 List the properties and uses of Nitric acid
- 2.2.2 Explain the reaction involved in the process
- 2.2.3 Draw the flow sheet for the manufacturing process
- 2.2.4 List the raw materials used in the process
- 2.2.5 State the catalyst used
- 2.2.6 Explain the construction and working of converter
- 2.2.7 Explain the heat recovery and utilization involved in the process
- 2.2.8 Explain the various grade of acid and concentration of acids

2.3.0 Understand the production of HCl acid

- 2.3.1 List the properties and uses of hydrochloric acid
- 2.3.2 List the raw materials.
- 2.3.3 Draw the flow diagram and explain the of manufacturing process

2.4.0 Understand the production of Phosphoric Acid

- 2.4.1 List the properties and uses
- 2.4.2 List the raw materials for the manufacturing process
- 2.4.3 Draw and explain the flow sheet for the manufacturing by wet process
- 2.4.4 Explain the overall reactions involved

MODULE – III

3.1.0 Understand the Manufacturing of Ammonia and Ammonium sulphate Fertilizer.

- 3.1.1 List the properties and uses of Ammonia
- 3.1.2 Explain the preparation of N_2 and H_2 from air and Naphtha with flow sheet
- 3.1.3 Explain and working of various equipments with above process
- 3.1.4 Explain the composition in the synthesis
- 3.1.5 Explain the catalytic conversion of Ammonia
- 3.1.6 Explain the working and constructional details of Ammonia converter
- 3.1.7 Explain the steam reforming process with flow sheet.

3.2.0 Understand the manufacture of ammonium sulphate and ammonium chloride

- 3.2.1 Explain the manufacturing process of ammonium sulphate by direct neutralization with gypsum with flow sheet.
- 3.2.2 List the equipments used in the above process.
- 3.2.3 Explain the manufacturing process of ammonium chloride by Dual salt process with flow sheet.

3.3.0 Understand the manufacturing of Urea

- 3.3.1 Explain the chemical reaction involved in the Urea manufacturing process.
- 3.3.2 Explain the kinetics and equilibria of the reaction involved
- 3.3.3 Explain the manufacturing process of Urea by total recycle process the flow diagram.

3.4.0 Understand the manufacture of Phosphatic fertilizers.

- 3.4.1 Describe manufacturing process of single super phosphate.
- 3.4.2 Describe the manufacturing process of Triple superphosphate.

MODULE – IV

4.1.0 Understand the manufacturing of pigments

- 4.1.1 List and explain various constituents of paints
- 4.1.2 Explain the manufacturing process of paints
- 4.1.3 Explain the manufacturing process of Titanium dioxide with flow sheet (Chloride process & Sulphate process)
- 4.1.4 Explain the manufacture of carbon black with flow sheet

4.2.0 Understand the Manufacturing of Sodium Silicate

- 4.2.1 List the raw materials for the manufacture of sodium silicate
- 4.2.2 Explain the manufacture of sodium silicate with the help of flow sheet
- 4.2.3 List the applications of sodium silicate

4.3.0 Understand the Manufacturing of Cement

- 4.3.1 List the raw materials of cement
- 4.3.2 Explain the dry process and wet process for the manufacture of cement with the help of flow sheet
- 4.3.3 Explain the construction and working of rotary drier
- 4.3.4 Explain the phenomenon of setting of cement

4.4.0 Understand the Manufacturing of Glass

- 4.4.1 List the raw material for the manufacture
- 4.4.2 Explain the manufacture of glass from raw materials
- 4.4.3 Explain the various grade of glass

- 4.4.4 List the properties and uses of various grade

4.5.0 Understand the Manufacturing of Refractory

- 4.5.1 List and explain various type of refractory
- 4.5.2 Classify various type of refractory
- 4.5.3 List the raw material for the manufacture
- 4.5.4 Explain the manufacture of various types

4.6.0 Understand the Manufacturing of Ceramics

- 4.6.1 List and explain various types of ceramics
- 4.6.2 List the properties and uses of various types

CONTENT DETAILS

MODULE I

ALKALIES

Preparation and purification of brine- reaction and flow diagram- light and heavy soda Ash - industrial applications- grades - Solvay and Modified Solvay process - Caustic Soda – raw materials – purification – membrane cells – manufacture of solid caustic soda - Chlorine – purification – liquefaction & uses

MODULE II

ACIDS

Sulphuric acid – Introduction – Importance of Sulphuric acid – properties and uses

Raw materials – Sulphur – pyrites – production of sulphur dioxide, sulphur burners –cleaning and cooling of gases – waste heat boilers. Contact process – theory of contact process – Equilibrium and rate equation – catalyst – vanadium pent oxide – platinum – flow diagram of entire process – working and constructional details of converter – absorber DCDA process with flow – sheet.

Nitric acid – Properties and uses – reaction – catalyst – manufacturing process –concentration of nitric acid grades. Hydrochloric Acid:- Properties and uses and commercial grades – manufacture of Hydrochloric acid by synthesis process with flow sheet.

Phosphoric Acid - Properties, uses and grades, raw materials manufacturing process.

Wet process

MODULE III

FERTILIZERS

Ammonia Synthesis – theoretical principles – equilibrium reaction - catalyst - space velocity.

Different source of synthesis gas low and high pressure process for the manufacture of Ammonia – properties and uses of Ammonia. Manufacture of ammonium sulphate – direct neutralization and gypsum process. – Manufacture of ammonium chloride by dual salt process.

Urea – Manufacture of urea – equilibrium and kinetics consideration – total recycle process – Manufacture of single super phosphate – Triple superphosphate.

MODULE IV

SILICATE INDUSTRIES

Pigments – TiO₂ manufacture with help of flow sheet – chloride process and sulphate process.

Carbon black – paints – constituents - Manufacture of sodium silicate – uses

cement – raw materials – main unit operation and process involved – wet and dry process

– merits & demerits. Glass – manufacturing method - different grades – uses.

Refractories - types – classification, properties – manufacture of fire clay brick, silicate brick, high alumina brick, carbon ceramics – raw material – main unit operation and process

REFERENCE

- | | | |
|------------------|---|----------------------------------|
| Dryden | - | Out lines of Chemical Technology |
| R. Norris Shreve | - | Chemical Process industries |
| T T T I, Chennai | - | Chemical Technology Volume - II |